UNIVERSITI TEKNOLOGI MARA
FINAL EXAMINATION

COURSE	PRE CALCULUS
COURSE CODE	MAT133
EXAMINATION	OCTOBER 2026
TIME	2 HOURS

NSTRUCTIONS TO CANDIDATES

1. This question paper consists of 5 questions.
2. Answer ALL questions in the Answer Booklet. Start each answer on a new page.
3. Do not bring any material into the examination room unless permission is given by the invigilator.
4. Please check to make sure that this examination pack consists of:
   i) the Question Paper
   ii) a one – page Appendix 1
   iii) an Answer Booklet - provided by the Faculty
   iv) a graph paper - provided by the Faculty
5. Answer ALL questions in English.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO

This examination paper consists of 6 printed pages

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QUESTION 1

a) Write the solution of 4 - 4x < -6 in interval notation and on a number line.

(3 marks)

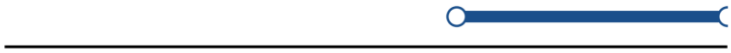


Figure: solution set.

b) Solve the inequality |2x - 2| < 4. Express the answer in interval notation and on a number line.

(5 marks)



Figure: solution on a number line.

c) Solve the equation |-2x - 3| = 8.

(4 marks)

QUESTION 2

a) Given $f(x) = (3 - 3x)/(4x + 4)$, $x \neq -1$, find $f^{-1}(x)$ and state its domain.

(5 marks)

b) The piecewise function is defined by
$f(x) = -2x - 1$, if $x < -1$;
$f(x) = 1$, if $-1 \leq x \leq 2$;
$f(x) = x^2 - 2$, if $x > 2$.
Sketch the graph and find f(-2), f(0) and f(3).

(5 marks)

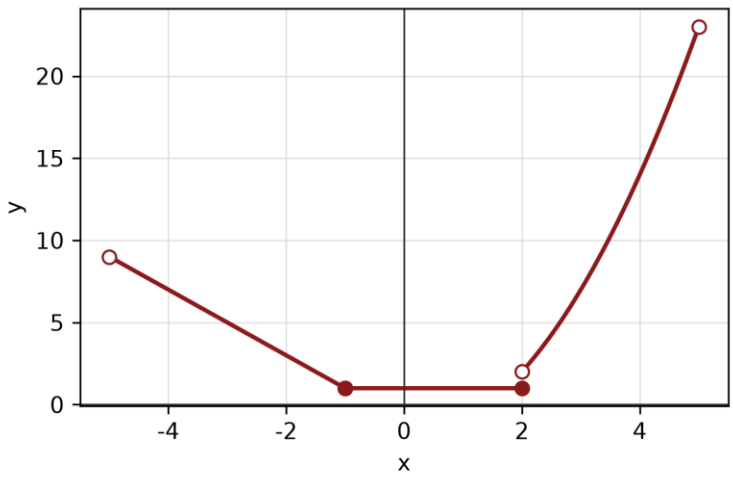


Figure: piecewise graph.

c) Sketch the circle $x^2 + 6x + y^2 - 8y = 0$. State its centre and radius.

(4 marks)

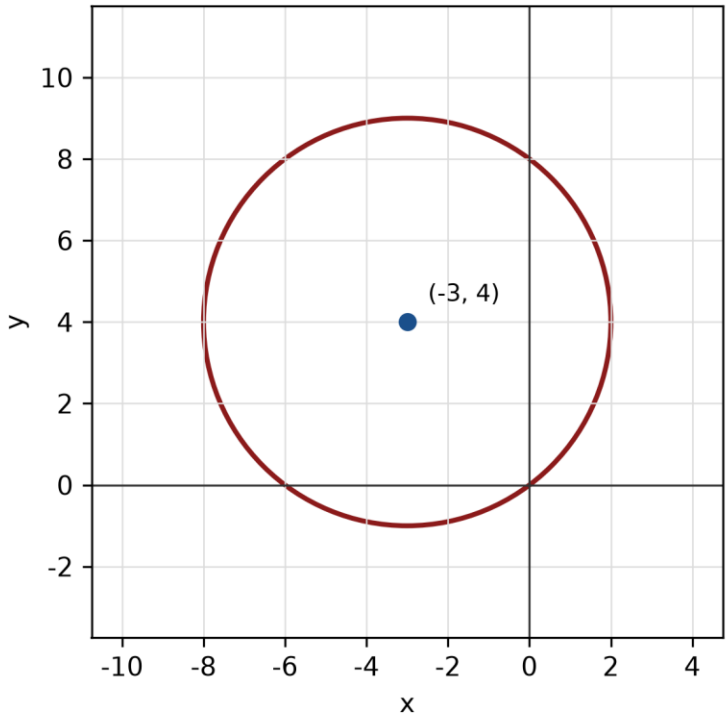


Figure: circle.

QUESTION 3

a) Given $z_1 = -5 + 3i$ and $z_2 = -1 + 5i$, express z_1/z_2 in the form $a + bi$.

(5 marks)

b) Find the modulus and argument of $z = -8 + 15i$. Give the argument in degrees correct to one decimal place.

(5 marks)

c) Use long division to find the quotient and remainder when $f(x) = -2x^3 + 5x^2 - 2x - 5$ is divided by x.

(5 marks)

QUESTION 4

a) Solve the simultaneous equations

$y = x^2 + 4$

$y = 7 - 2x$.

(5 marks)

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b) Write a system of inequalities that describes the shaded triangular region with vertices (0, 0), (4, 0) and (0, 5).

(5 marks)

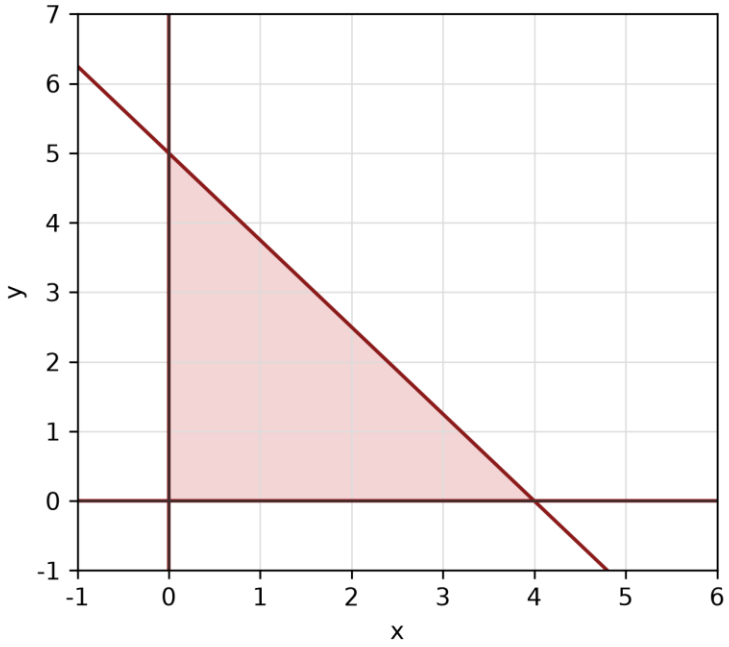


Figure: shaded triangular region.

c) Write the equation of the circle with centre (-1, 1) and radius 2. Hence write its general equation and sketch the circle.

(5 marks)

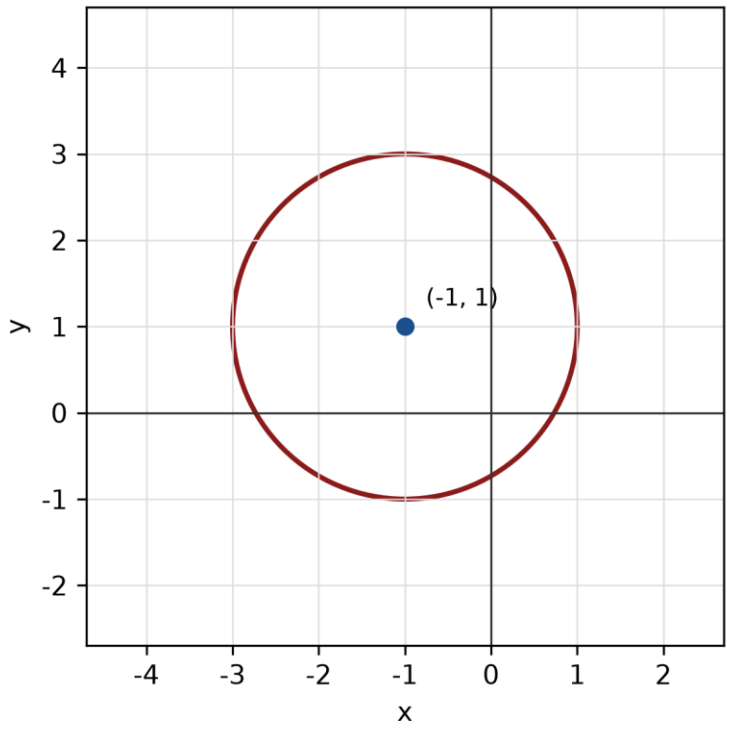


Figure: circle.

QUESTION 5

a) Given sin A = 5/13 where A is in quadrant II and cos B = 12/13 where B is in quadrant IV,

find sin 2A and cos(A + B).

(5 marks)

b) Find the perimeter of the sector with radius 11 cm and central angle 45°.

(4 marks)

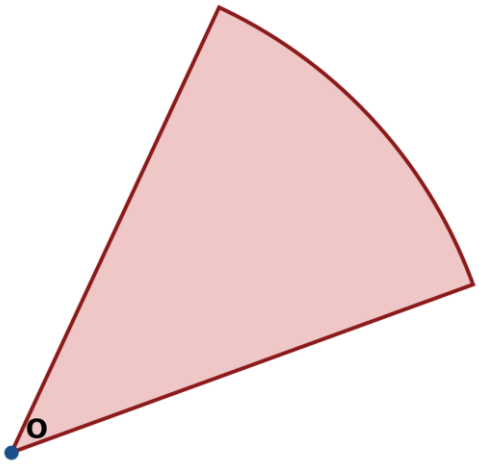


Figure: sector.

c) In triangle ABC, b = 8 cm, c = 12 cm and A = 60°. Use the cosine rule to find a.

(4 marks)

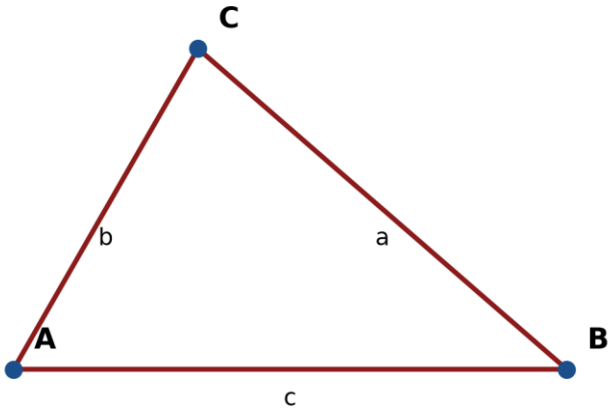


Figure: triangle ABC.

END OF QUESTION PAPER

APPENDIX 1

Formula sheet (as supplied with the paper).

$\sin^2 A + \cos^2 A = 1$
$\sin(A + B) = \sin A \cos B + \cos A \sin B$
$\cos(A + B) = \cos A \cos B - \sin A \sin B$
$\tan(A - B) = (\tan A - \tan B) / (1 + \tan A \tan B)$
$\cos 2A = \cos^2 A - \sin^2 A$
$\cos 2A = 1 - 2\sin^2 A = 2\cos^2 A - 1$
$\sin 2A = 2 \sin A \cos A$
$\tan^2 A + 1 = \sec^2 A$
$\tan 2A = 2 \tan A / (1 - \tan^2 A)$
Arc length, $s = r\theta$
Area of sector = $\frac{1}{2}r^2\theta$
$a / \sin A = b / \sin B = c / \sin C$
$a^2 = b^2 + c^2 - 2bc \cos A$
Area of triangle = $\frac{1}{2}ab \sin C$

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